Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	16 · 559 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Split / seed / CV	random_state=42 · CV: LOUO (16 folds, every user held out once)
Sample counts	train (post-aug)=559 · LOUO-evaluated samples=559

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 16 folds)	CV std	CV min	CV max
Random Forest	0.9306	0.0721	0.7714	1.0000
Logistic Regression	0.9288	0.1065	0.5714	1.0000
LDA	0.9122	0.0960	0.6857	1.0000
SVM (Linear)	0.9078	0.1124	0.6286	1.0000
SVM (RBF)	0.9022	0.1209	0.5714	1.0000
KNN	0.7394	0.0804	0.6000	0.8857

Per-class CV F1 (across all LOUO folds)

Class	Random Forest	Logistic Regression	LDA	SVM (Linear)	SVM (RBF)	KNN	Support
Double_Closed_Fist	0.936	0.945	0.877	0.951	0.933	0.693	83
Double_Nothing	0.766	0.795	0.794	0.750	0.750	0.541	75
Double_Okay	0.994	0.975	0.981	0.923	0.896	0.683	80
Double_Open_Palm	0.982	0.924	0.940	0.936	0.923	0.700	80
Double_Phone	0.928	0.949	0.886	0.901	0.909	0.587	80
Double_Pistol	0.902	0.919	0.909	0.905	0.916	0.920	81
Double_Spiderman	0.982	0.982	0.981	0.982	0.982	0.947	80
macro avg	**0.927**	**0.927**	**0.910**	**0.907**	**0.901**	**0.724**	**559**

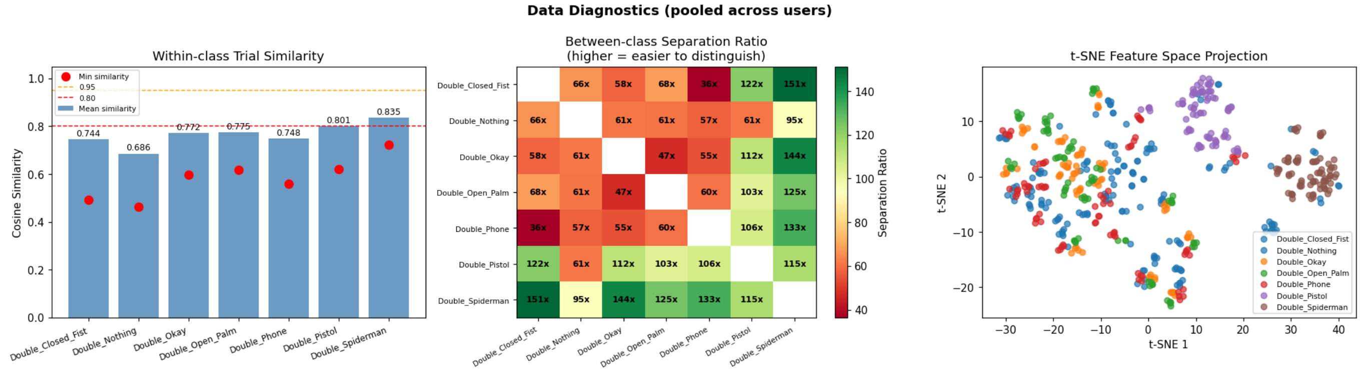
Per-user accuracy (each user as held-out fold)

User	n trials	Random Forest	Logistic Regression	LDA	SVM (Linear)	SVM (RBF)	KNN
1_Alan	35	1.000	1.000	1.000	0.971	0.971	0.629
2_Alex	35	0.771	0.829	0.714	0.686	0.714	0.686
3_Anghad	35	0.886	0.829	0.857	0.857	0.829	0.714
4_Daniel	35	0.943	0.971	0.886	0.971	0.971	0.743
5_Georgia	35	1.000	1.000	1.000	1.000	1.000	0.886
6_Harry	35	0.857	0.943	0.943	0.914	0.914	0.771
7_Henry	35	0.829	0.943	0.686	0.800	0.743	0.743
8_Jestin	36	1.000	0.944	1.000	0.972	0.972	0.694
9_Joselyn	30	1.000	1.000	0.933	1.000	1.000	0.800
10_Josh	35	0.857	0.571	0.829	0.629	0.571	0.629
11_Mansh	36	1.000	0.972	0.944	0.972	0.944	0.833
12_Marcus	35	1.000	1.000	1.000	1.000	1.000	0.857
13_Nat	37	0.919	0.973	0.946	0.838	0.919	0.703
14_StephenV2	35	0.971	1.000	1.000	1.000	1.000	0.600
16_Bella	35	0.886	0.914	0.886	0.943	0.943	0.743
17_Raquel	35	0.971	0.971	0.971	0.971	0.943	0.800
Mean (across users)		0.931	0.929	0.912	0.908	0.902	0.739
Std (across users)		**0.072**	**0.106**	**0.096**	**0.112**	**0.121**	**0.080**

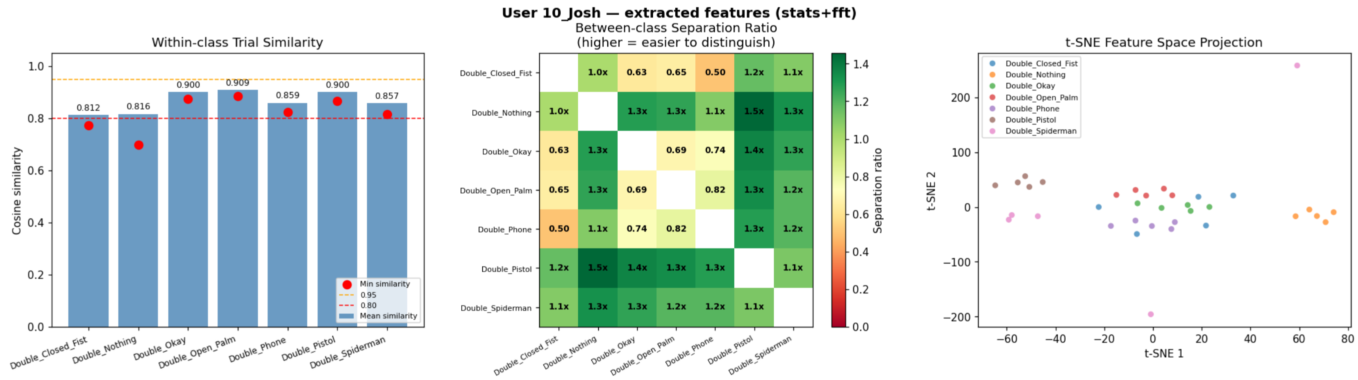
Appendix — Data Diagnostics

17 figures: pooled across users + one per detected user (extracted-feature views).

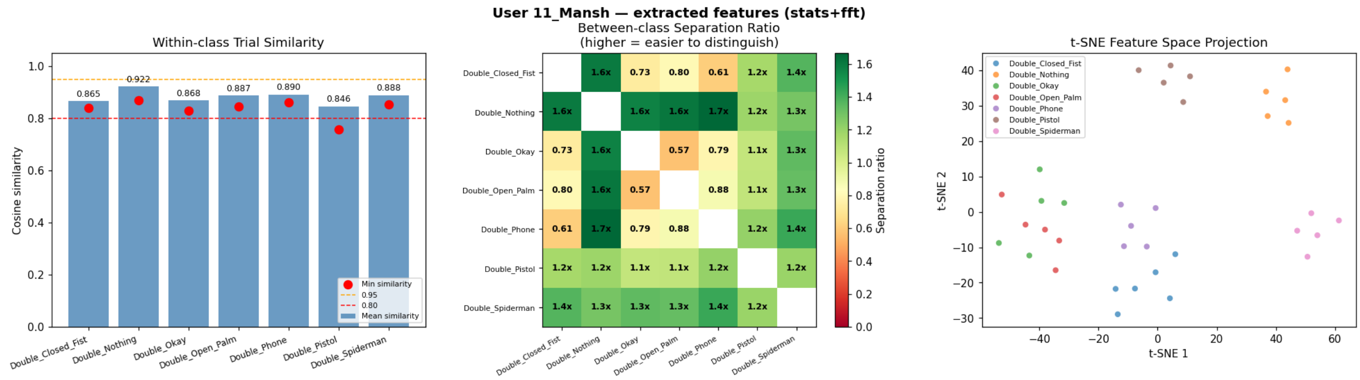
Pooled diagnostics — extracted features (across all users)



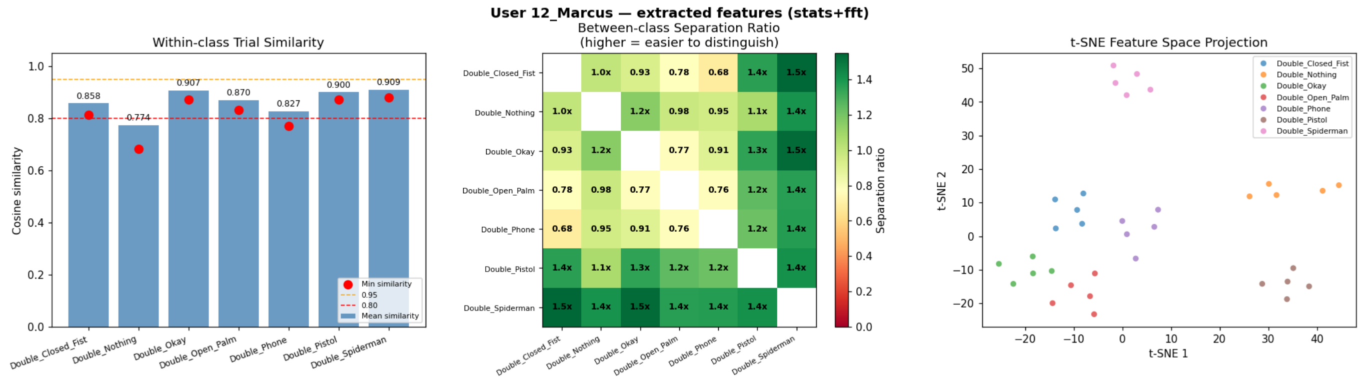
User 10_Josh — extracted features (stats+fft)



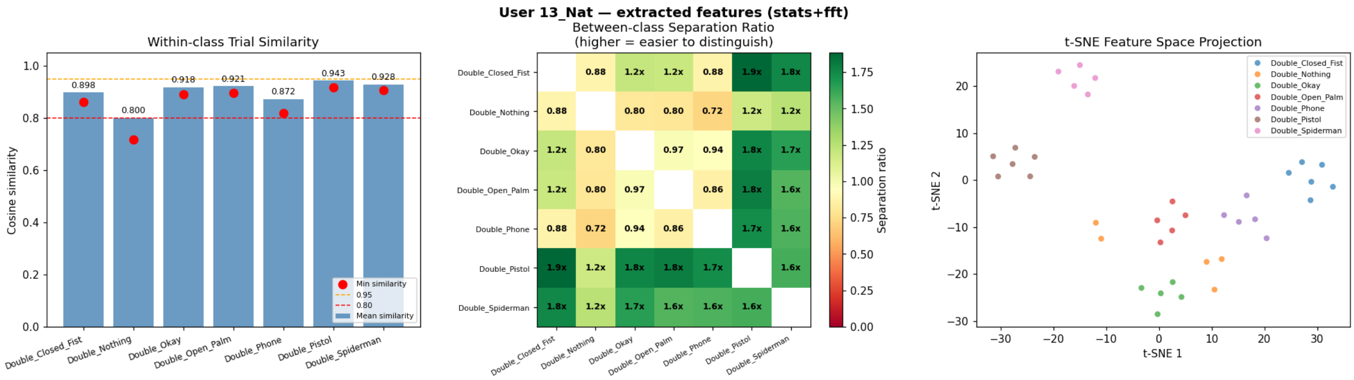
User 11_Mansh — extracted features (stats+fft)



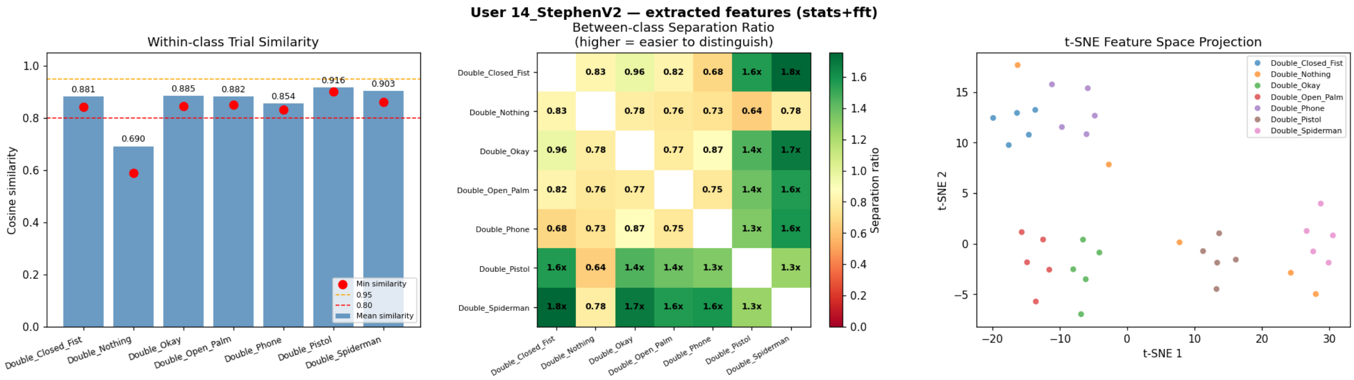
User 12_Marcus — extracted features (stats+fft)



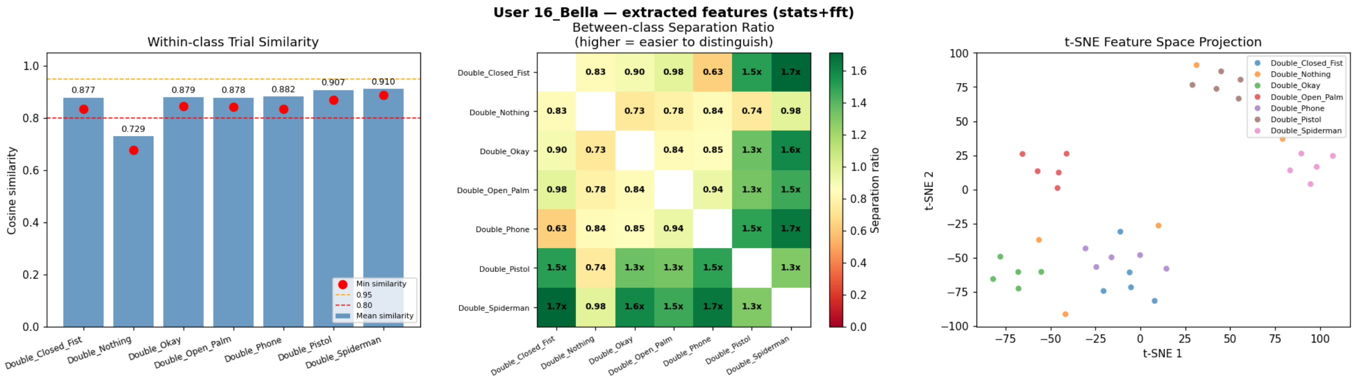
User 13_Nat — extracted features (stats+fft)



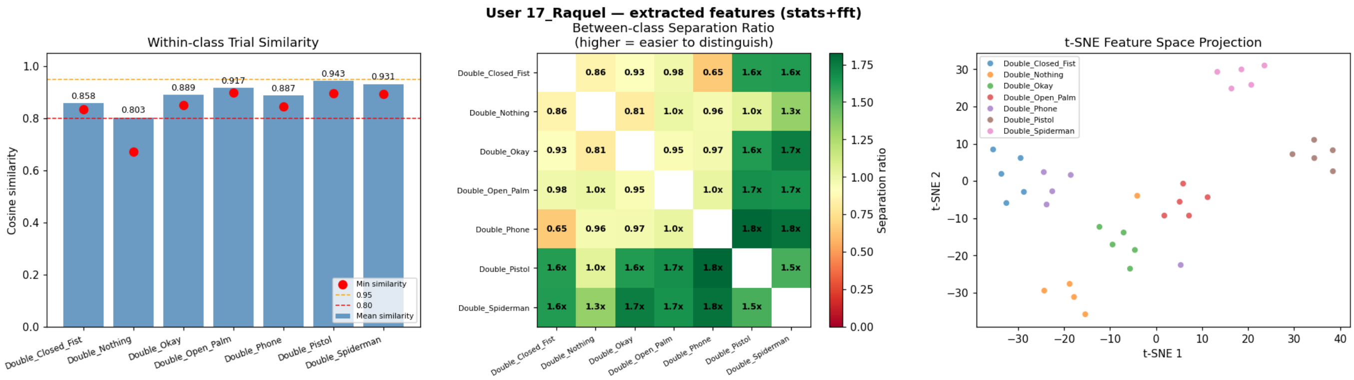
User 14_StephenV2 — extracted features (stats+fft)



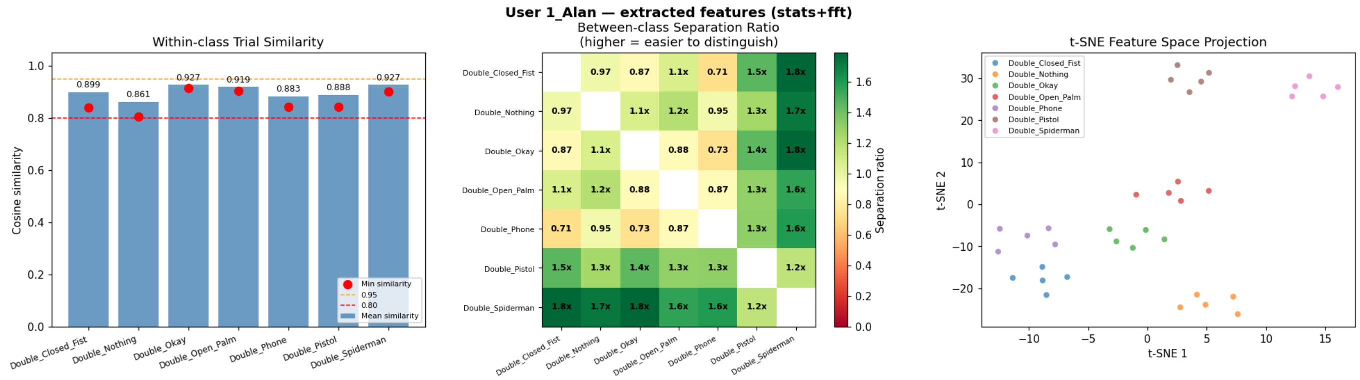
User 16_Bella — extracted features (stats+fft)



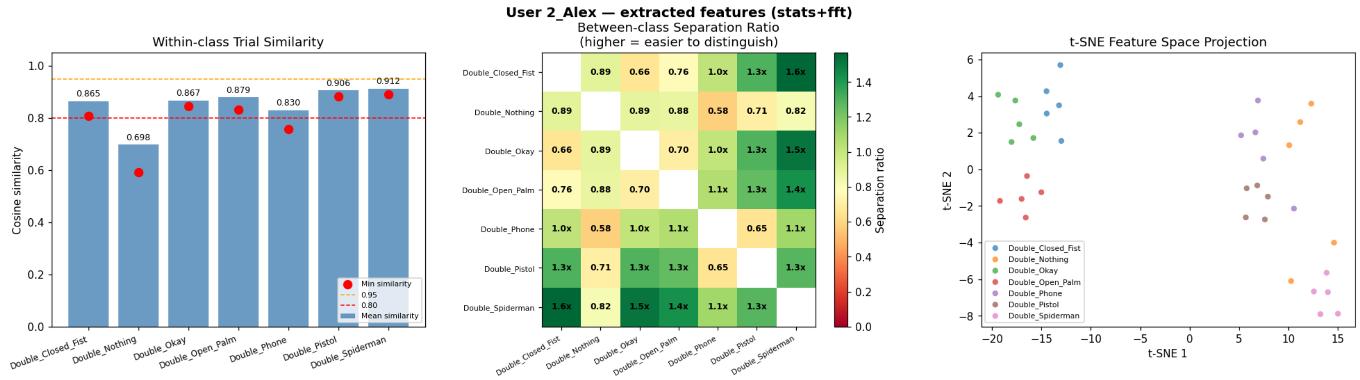
User 17_Raquel — extracted features (stats+fft)



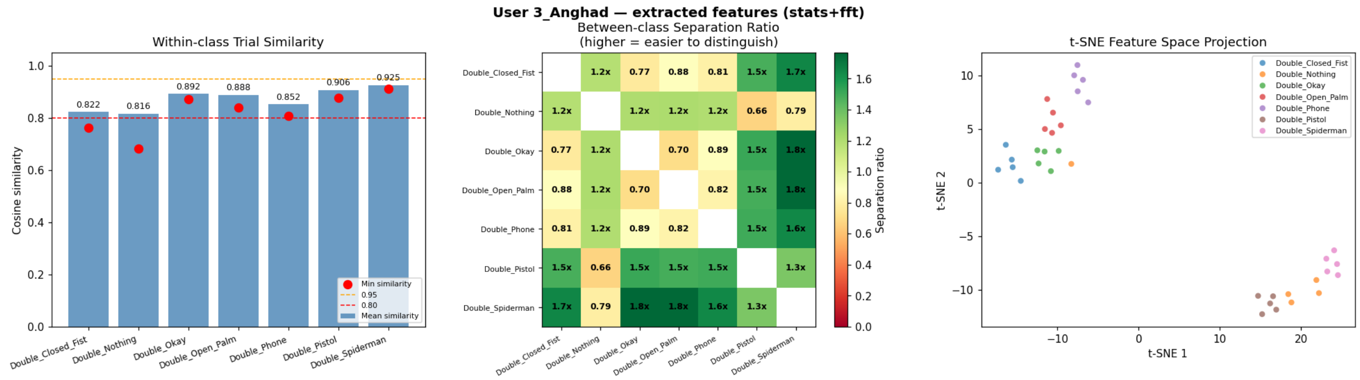
User 1_Alan — extracted features (stats+fft)



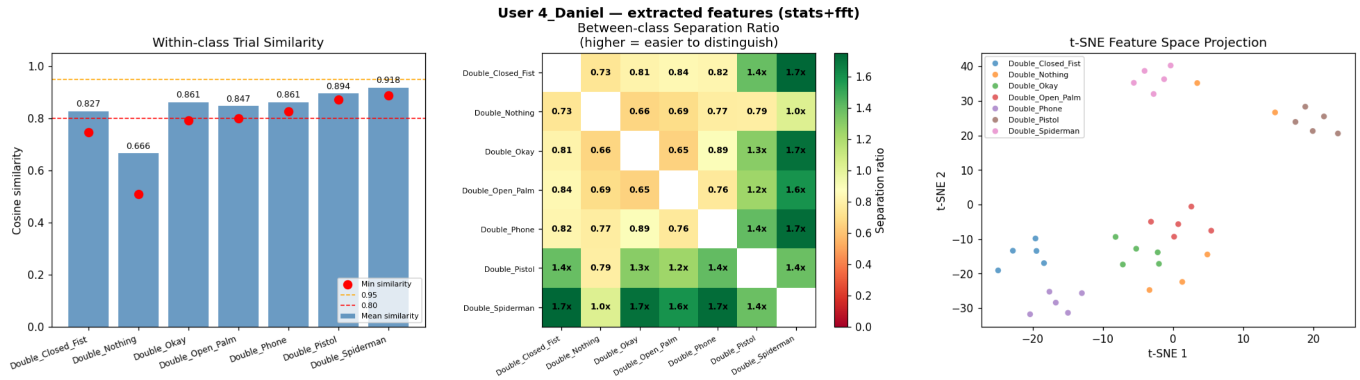
User 2_Alex — extracted features (stats+fft)



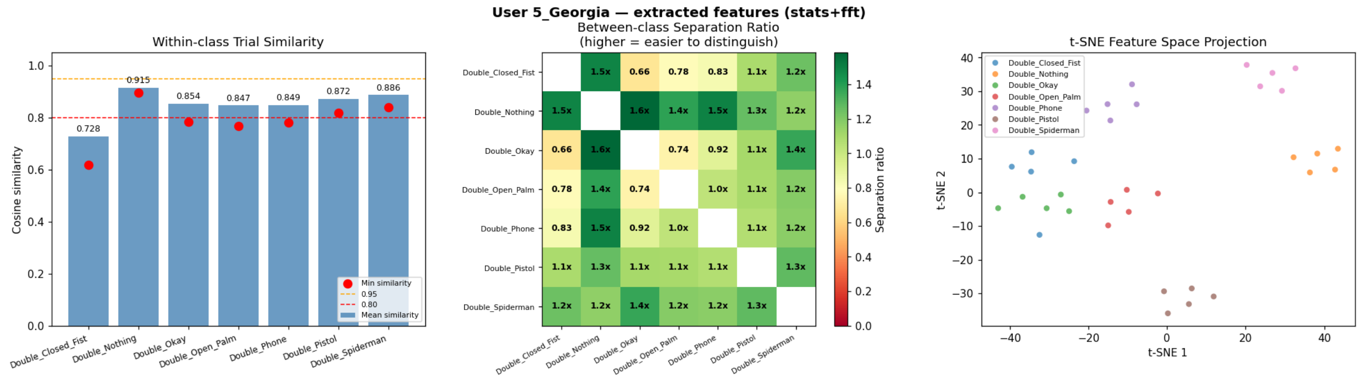
User 3_Anghad — extracted features (stats+fft)



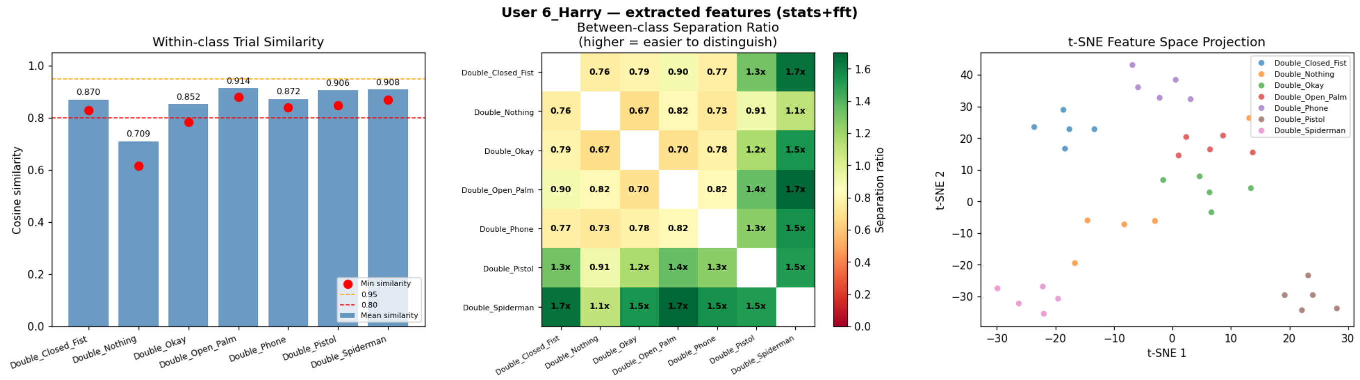
User 4_Daniel — extracted features (stats+fft)



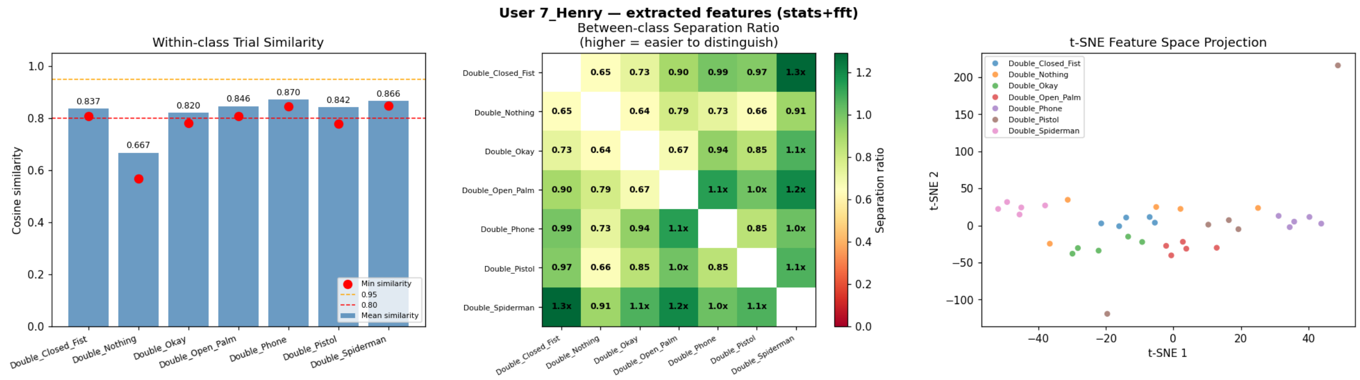
User 5_Georgia — extracted features (stats+fft)



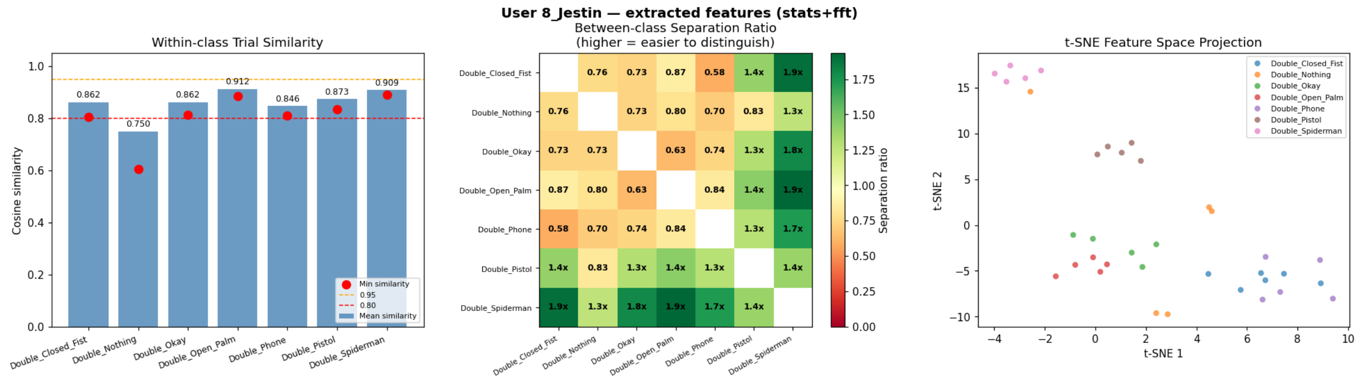
User 6_Harry — extracted features (stats+fft)



User 7_Henry — extracted features (stats+fft)



User 8_Jestin — extracted features (stats+fft)



User 9_Joselyn — extracted features (stats+fft)

User 9_Joselyn — extracted features (stats+fft)